

1. Plan Your Database – an Online Bookstore Database

- **My scenario:** An online bookstore sells books to customers. Customers place orders, and each order may include multiple books.
- **Tables:**

Customers

Books

Orders

Order_Items

2. Define Tables and **Primary Keys:**

Customers

- **'customer_id'**
- 'first_name'
- 'last_name'
- 'email_address'
- 'address'
- 'city'
- 'zip_code'

Orders

- **'order_id'**
- 'order_date'
- 'customer_id'

Order_Items

- **'order_item_id'**
- 'order_id'
- 'book_id'
- 'quantity_book'
- 'unit_price'

Books

- **'book_id'**
- 'title_book'
- 'author'
- 'date_publication'
- 'price'
- 'stock_quantity'

3. Show Relationships and Cardinality:

Cardinality:

- Customers > Orders – one customer can place many orders. **Cardinality = one to many**
- Orders > Order_Items – one order can contain many items. **Cardinality = one to many**
- Orders >< Books – one order can contain many book and one book_id can go to many orders. **Cardinality = many to many**
- Book > Order_item – one book can go in many order items. **Cardinality = one to many**

4. Normalise Your Data:

First Normal Form (1NF)

- Each table has a primary key
- Each field contains a single value
- No lists or repeating groups

Example: Books in an order are stored as separate rows in Order_Items table.

Second Normal Form (2NF)

- All non-key fields depend on the full primary key

Example: Customer details are stored only in Customers table. Book details are stored only in Books table.

Third Normal Form (3NF)

- No non-key field depends on another non-key field

Example: Customer email depends only on 'customer_id'. Book price depends only on 'book_id'.

5. Create Your Diagram (primary key, foreign key, cardinality):

